

Angular Reference

TERMS • ANALOGIES • TECHNICAL DEFINITIONS

Core
Framework
RxJS / Async
HTTP /
Routing
State (NgRx)
Meta /
Runtime

CORE FRAMEWORK		
TERM	ANALOGY	TECHNICAL DEFINITION
Angular	<i>The building's architecture blueprint</i>	Frontend framework by Google for building SPAs
Component	<i>A room in the building</i>	UI building block – class + template + styles
Template	<i>The room's interior design</i>	HTML with Angular syntax bound to the component class
Service	<i>Building plumbing – shared, runs behind walls</i>	Injectable singleton class for shared logic/data
Dependency Injection	<i>Calling reception instead of building it yourself</i>	Design pattern – Angular provides class instances via inject()
Standalone	<i>A room that brings its own furniture</i>	Component that declares its own imports, no NgModule needed
Module (legacy)	<i>A floor grouping related rooms</i>	NgModule – legacy grouping mechanism for components/services
Decorator	<i>A name badge clipped onto a class</i>	TypeScript metadata annotation – e.g. @Component, @Injectable
Input / Output	<i>Room door slot – data in, events out</i>	Component API – input() passes data down, output() emits events up
Lifecycle Hook	<i>Room alarm at key moments – entering, settling, leaving</i>	Interface methods Angular calls at defined points in a component's life

SIGNALS & CHANGE DETECTION		
TERM	ANALOGY	TECHNICAL DEFINITION
Signal	<i>A live scoreboard – updates the exact screen watching</i>	Reactive primitive – tracked value that notifies dependants on change
Change Detection	<i>Building inspector walking every room</i>	Mechanism that syncs component state to the DOM

OnPush	<i>Inspector only visits if you call him</i>	CD strategy – skips component unless input ref changes or event fires
Zone.js	<i>Spy bug on every phone – reports when any call ends</i>	Third-party library Angular uses to intercept async APIs
Monkey Patch	<i>Secretly swapping the office phone with a bugged version</i>	Runtime override of a native browser API without caller knowing

RXJS / ASYNC

TERM	ANALOGY	TECHNICAL DEFINITION
Observable	<i>A news subscription – delivers whenever there's a story</i>	RxJS lazy data stream that emits values over time
Subscribe	<i>Actually opening your mailbox</i>	Method that starts execution of an Observable
Pipe	<i>Conveyor belt with filters before package reaches you</i>	RxJS .pipe() – chains operators to transform a stream
switchMap	<i>Cancelling your Uber and ordering a new one</i>	RxJS operator – cancels previous inner Observable, starts new one
tap	<i>Security camera – watches but doesn't touch</i>	RxJS operator – side effect only, passes value through unchanged
map	<i>Factory worker reshaping every item on the belt</i>	RxJS operator – transforms each emitted value
Async Pipe	<i>Butler who opens mail, reads it, cleans up after</i>	Angular pipe that subscribes, unwraps, and auto-unsubscribes

HTTP / ROUTING

TERM	ANALOGY	TECHNICAL DEFINITION
Router	<i>Building directory – tells you which room to go to</i>	Angular module managing client-side navigation via URL
Guard	<i>Bouncer at the room entrance</i>	Functional route guard – controls navigation access
Interceptor	<i>Customs – inspects every package in and out</i>	HTTP middleware – intercepts requests and responses globally

NGRX / STATE

TERM	ANALOGY	TECHNICAL DEFINITION
NgRx Store	<i>Single whiteboard everyone in the building reads from</i>	Redux-pattern global state container for Angular

Selector	<i>Highlighter on the whiteboard — shows only your section</i>	Pure function that derives a slice of state from the Store
Effect	<i>Task triggered when someone updates the whiteboard</i>	NgRx side effect handler — listens for actions, runs async logic